Shift Registers

Tutorial 7 - Digital Design (CS F215)

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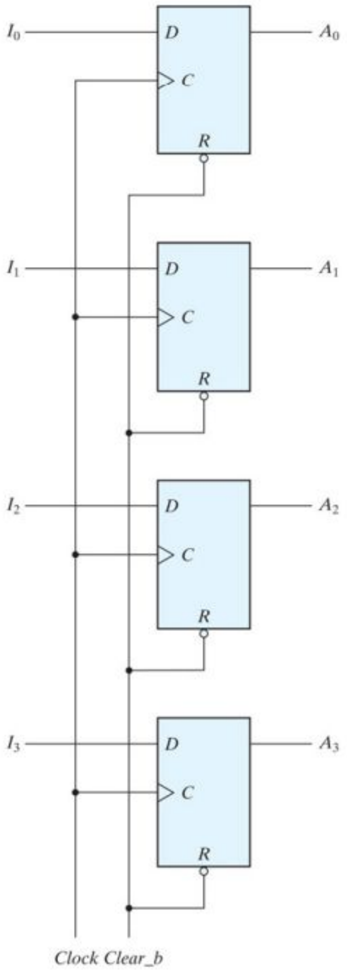
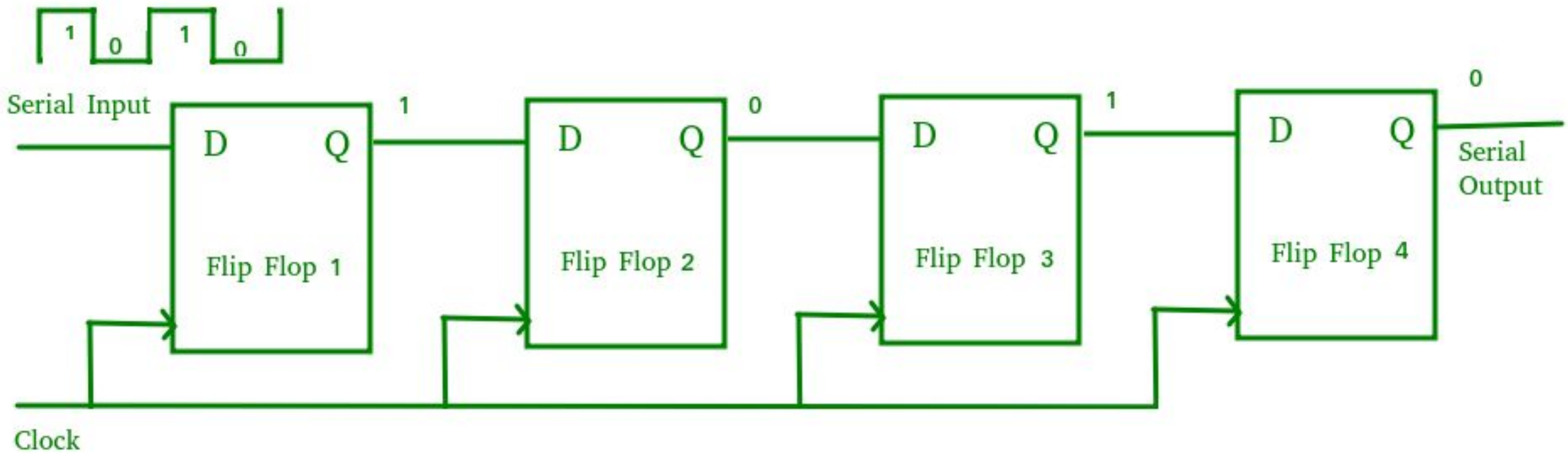
What is a Register?

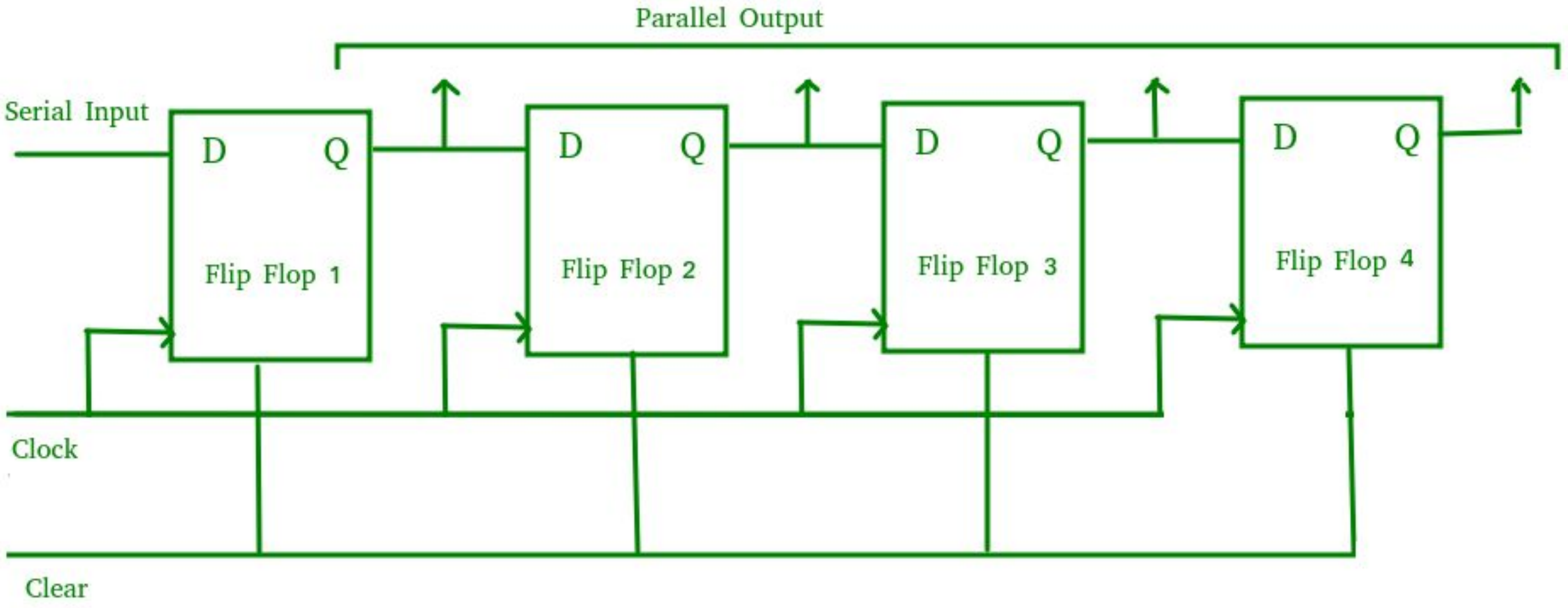
- A register is a group of flip-flops sharing a common clock.
- Each flip-flop in a register can store one bit of information.
- An n-bit register consists of a group of n flip-flops capable of storing n bits of binary information.



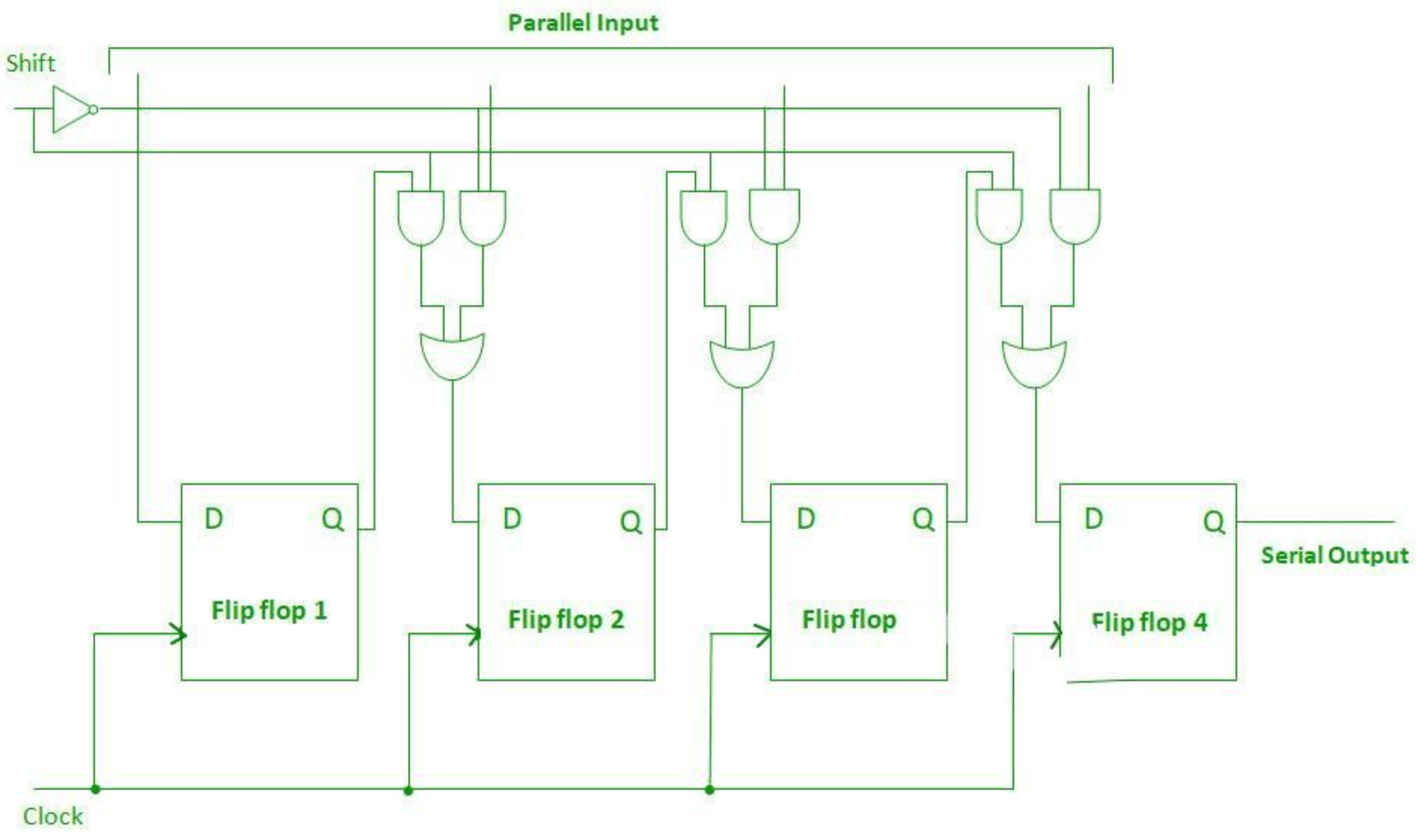
Serial-In Serial-Out Shift Register (SISO)



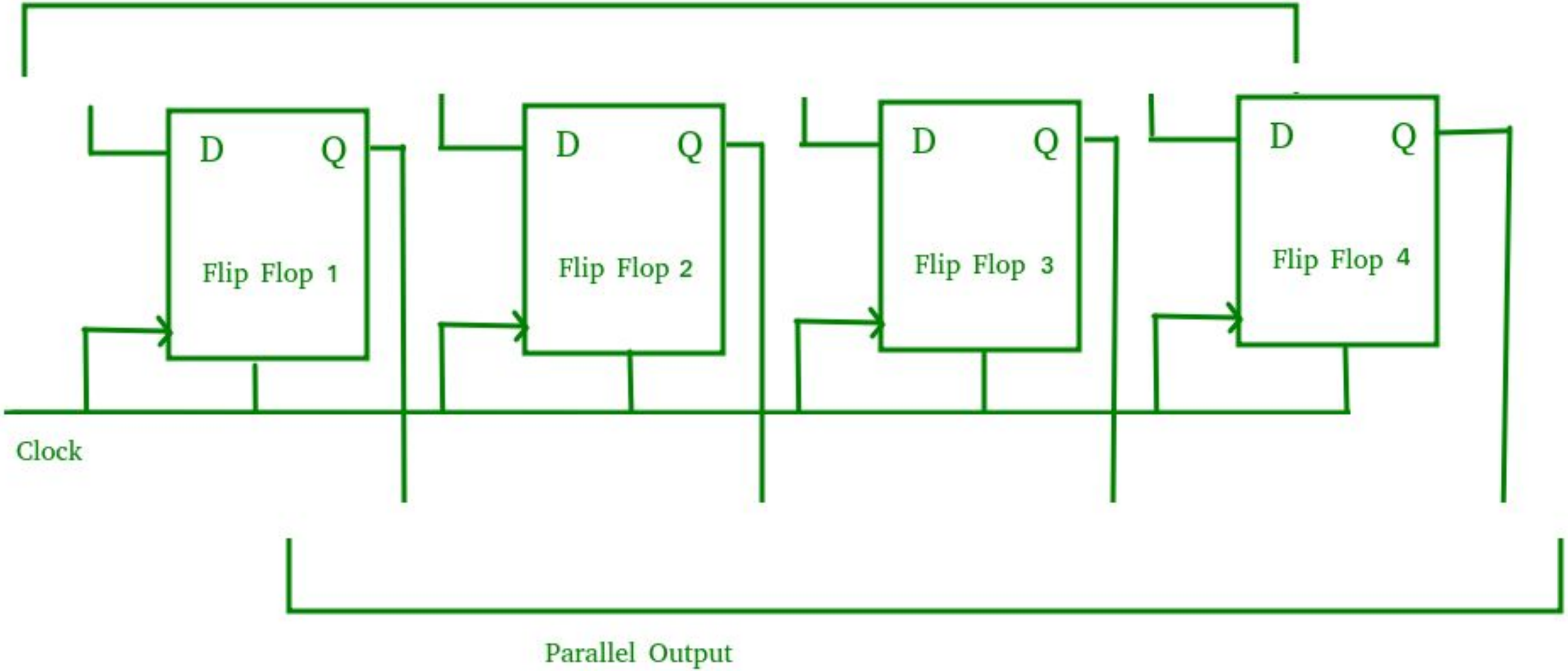
Serial-In Parallel-Out Shift Register (SIPO)



Parallel-In Serial-Out Shift Register (PISO)



Parallel-In Parallel-Out Shift Register (PIPO)



Q. Design a **pattern detector** that identifies the bit sequence `1011` from a serial input stream using a **shift register** and combinational logic.

1. **Shift Register Module**:
   - Store the last 4 bits of a serial input.
   - Shift left on each clock cycle and insert the new bit at LSB.

2. **Pattern Detector Module**:
   - Read the shift register value.
   - Output `1` when the pattern `1011` is detected, else `0`.

Note: The **top module** and **testbench** have already been implemented.

Example:

`Input Bitstream: 1 0 1 1 0 1 1`

`Expected Detection Output: 0 0 0 1 0 0 1`

- Each `1` in the output shows that `1011` was detected at that clock cycle.

Thank You